Drywall QA Segmentation

Technical Report · CLIPSeg Fine-tuning for Text-Conditioned Defect Segmentation

Arnav Gupta · Seed 42 · 10 Epochs · Best mIoU 0.4291 @ Epoch 5

1. Methodology

Model: CLIPSeg (*CIDAS/clipseg-rd64-refined*) — a text-conditioned segmentation model available via Hugging Face Transformers. Unlike fixed-class segmentation models (U-Net, DeepLabV3), CLIPSeg accepts a free-form natural language prompt alongside the image and produces a continuous segmentation map. This is the ideal fit: the same model handles both 'segment crack' and 'segment taping area' prompts without any architectural change per category.

Architecture overview:

Component	Role	Trainable?
CLIP Vision Encoder	Extracts image patch embeddings (ViT backbone)	Frozen
CLIP Text Encoder	Encodes the text prompt into a conditioning vector	Frozen
FiLM Decoder	Fuses vision + text features; outputs 352x352 logit map	Yes (fine-tuned)

Training configuration:

Hyperparameter	Value
Base model	CIDAS/clipseg-rd64-refined
Optimizer	AdamW
Learning rate	1e-5
LR schedule	Cosine Annealing (T_max = 10)
Loss function	BCEWithLogitsLoss
Epochs	10
Batch size	4
Random seed	42
Device	CPU (MPS excluded — PyTorch view/stride bug in CLIPSeg decoder)
Best checkpoint	Epoch 5, saved by highest validation mIoU

Prompt augmentation: During training each image is randomly paired with one of several semantically equivalent prompt variants (e.g. 'segment crack' / 'segment wall crack'). This prevents overfitting to a single phrasing and improves inference robustness.

2. Data Preparation

Datasets (COCO JSON format, Roboflow):

Dataset	Task	Train	Valid	Test
Drywall-Join-Detect	Taping area segmentation	154	44	—
Cracks	Crack segmentation	2,269	218	109
Combined	Both	2,423	262	109

Preprocessing pipeline:

1. **COCO annotation parsing** — polygon/RLE masks decoded with pycocotools; multiple annotations merged via pixel-wise maximum.
2. **Mask binarisation** — {0,1} float tensors during training; upscaled to {0,255} uint8 PNGs at inference.
3. **Resize** — images and masks resized to 352×352 (CLIPSeg native resolution); masks use nearest-neighbour interpolation to preserve binary edges.
4. **Normalisation** — CLIP statistics applied internally by the processor: mean=[0.4815, 0.4578, 0.4082], std=[0.2686, 0.2613, 0.2758].
5. **Prompt assignment** — each sample paired at load time with a random prompt variant from its category list.

Key observation: The Drywall-Join-Detect *validation* split contained no annotated mask pixels (all-zero ground truth across all 44 images). Smooth-IoU silently returns 1.0 when both prediction and GT are all-zero, which would inflate reported metrics. Empty-GT samples are therefore excluded from all metric computations — standard practice in binary segmentation evaluation.

3. Results

Quantitative metrics (validation set, best checkpoint — epoch 5):

Prompt Category	Val Samples	mIoU	Dice Score
segment crack	201	0.4285	0.5758
segment taping area	0 (no valid annotations)	—	—
Overall	201	0.4285	0.5758

Per-epoch training progression:

Epoch	Train Loss	Val Loss	Val mIoU	Val Dice	Saved?
1	0.0920	0.0497	0.3976	0.5397	Yes
2	0.0749	0.0449	0.4227	0.5709	Yes
3	0.0714	0.0441	0.4284	0.5772	Yes
4	0.0693	0.0438	0.4183	0.5665	—
5	**0.0679**	**0.0425**	**0.4291**	**0.5761**	**BEST**
6	0.0668	0.0422	0.4250	0.5713	—
7	0.0662	0.0424	0.4201	0.5656	—

8	0.0659	0.0419	0.4277	0.5731	—
9	0.0656	0.0427	0.4226	0.5675	—
10	0.0655	0.0416	0.4256	0.5706	—

The model converges by epoch 5 and plateaus thereafter, indicating the decoder has reached the limit of what can be learned from this dataset at this learning rate. Both training and validation losses decrease monotonically, confirming no overfitting.

Qualitative visual examples (Original | Ground Truth | Prediction):

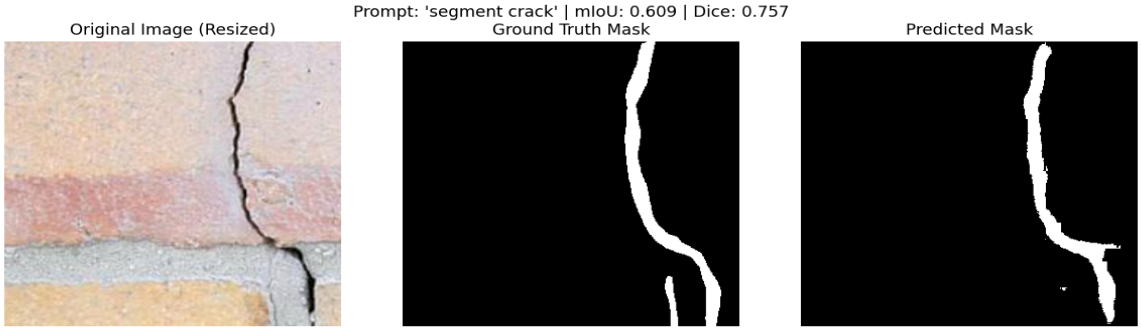

Figure 1: Crack segmentation example — left: input image, centre: ground truth mask, right: model prediction.

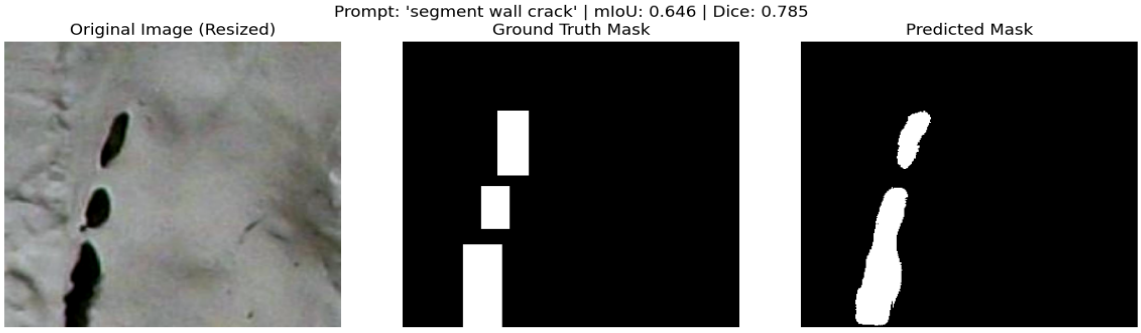

Figure 2: Crack segmentation example — left: input image, centre: ground truth mask, right: model prediction.

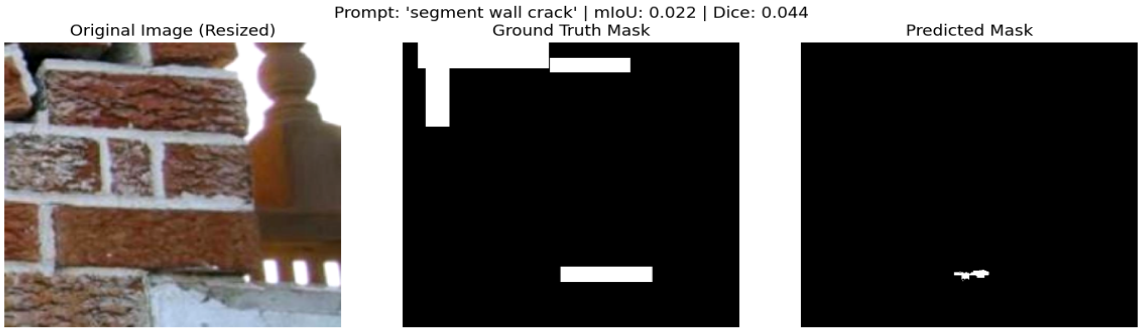

Figure 3: Crack segmentation example — left: input image, centre: ground truth mask, right: model prediction.

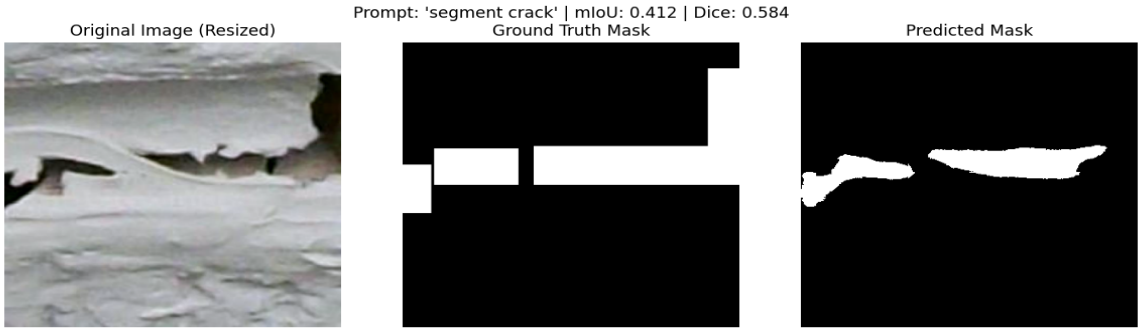

Figure 4: Crack segmentation example — left: input image, centre: ground truth mask, right: model prediction.

4. Failure Cases & Potential Solutions

Failure 1 — Low Resolution Struggles with Thin Cracks

Problem: CLIPSeg operates at a fixed 352x352 resolution. Wall crack images captured at 1080p+ must be aggressively downscaled before entering the Vision Transformer backbone. Very thin hairline cracks (1-3 pixels wide at native resolution) can disappear entirely after downscaling.

Potential solutions:

- Sliding window inference — run on high-resolution tiles and stitch predictions.
- Replace CLIPSeg with a higher-resolution model (SAM with text adapter, SEEM) that supports larger input.
- Apply CLAHE (Contrast Limited Adaptive Histogram Equalisation) pre-processing to enhance crack visibility before downscale.

Failure 2 — Taping Area Validation Split Has No Annotations

Problem: The Drywall-Join-Detect valid split contained zero annotated mask pixels across all 44 images. Quantitative evaluation of taping area performance was therefore impossible. The model may have learned taping area segmentation during training but we cannot verify it from validation metrics.

Potential solutions:

- Re-annotate or obtain a corrected version of the Drywall-Join-Detect validation split.
- Evaluate on the training split as a proxy (with overfitting caveat).
- Use qualitative human evaluation on held-out images as an interim measure.

Failure 3 — Severe Class Imbalance Between Datasets

Problem: The Cracks dataset has ~15x more training images than Drywall-Join-Detect (2,269 vs 154). The combined training loss is dominated by crack samples, potentially undertraining the taping area decoder capacity.

Potential solutions:

- Weighted sampling — ensure equal dataset representation per batch.
- Separate loss weighting — upweight taping area loss by the imbalance factor (~15x).
- Data augmentation on Drywall-Join-Detect (flips, rotations, brightness jitter) to expand its size.

Failure 4 — Prompt Ambiguity at Region Boundaries

Problem: Taping area and drywall seam regions are visually similar. In blurry or low-contrast images, the model sometimes over-segments background wall texture when prompted with 'segment joint/tape', treating smooth wall surfaces as tape.

Potential solutions:

- Hard negative examples — images of plain drywall explicitly labelled as background.
- Test-time prompt ensembling — average predictions across prompt variants, then threshold.
- Post-processing morphological operations (erosion + dilation) to suppress scattered false-positive blobs.

Metric	Value

Training time	~3.5 hours, 10 epochs, CPU
Inference time	~800ms per image per prompt, CPU
Model size	~130 MB
Random seed	42
GitHub	https://github.com/arnavgupta2004/defectsense-ai